CS341 · COMPUTER GRAPHICS

Calculations Practice

Practice on every calculation type

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01 Questions

Q1 ASPECT RATIO

a) A screen is 12 inches wide and 9 inches high with a resolution of 1024 × 768. Find the aspect ratio.

b) A screen is 10 inches wide and 8 inches high with a resolution of 800 × 600. Find the aspect ratio.

Q2 RESOLUTION

A monitor measures 32 cm × 24 cm and shows 1280 × 960 pixels. Find the resolution in pixels per cm in x and y, and the aspect ratio.

Q3 FRAME BUFFER

How much frame-buffer memory (in bytes) does a 1024 × 768 screen need with:
a) 1 bit per pixel b) 8 bits per pixel c) 24 bits per pixel

Q4 VECTORS

a) P = (2, 5), Q = (7, 1). Find the vector from P to Q and its length.

b) a = (1, 2, 2). Find |a| and the unit vector â.

c) b = (2, −1, 0). Find a · b. Are a and b perpendicular?

d) Find a × b.

e) Find the angle between u = (1, 0) and w = (−1, 1).

f) Find the counterclockwise perpendicular of c = (3, −2).

Q5 POINTS & LINES

a) Which are valid points? $0.5P + 0.5R$, $2P - R$, $P + R$, $0.2P + 0.3R$

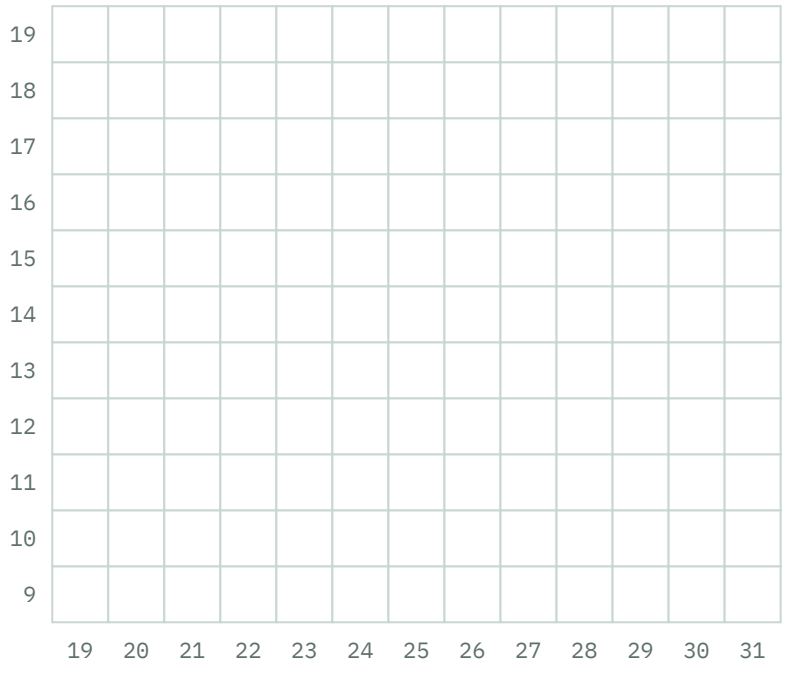
b) $A = (1, 2)$, $B = (9, 6)$. Find the point at $t = 0.75$ on $P = A + t(B - A)$.

c) For $C = (2, 1)$ and $B = (5, 5)$, write the parametric form and the point-normal form of the line.

Q6 DDA

Using DDA, find the pixels from (20, 10) to (30, 18).

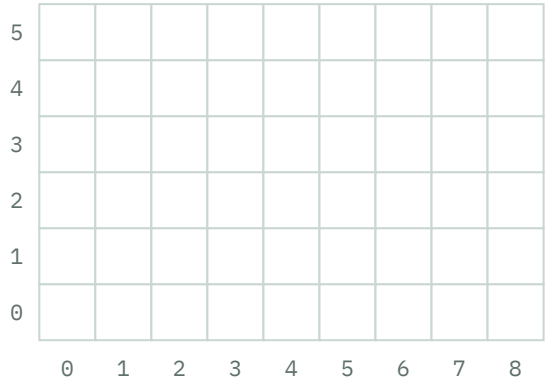
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				



Q7 DDA

Using DDA, find the pixels from (1, 1) to (7, 4).

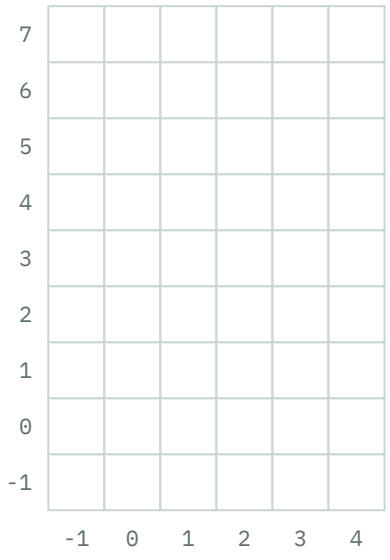
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				



Q8 DDA · |m| > 1

Using DDA, find the pixels from (0, 0) to (3, 6).

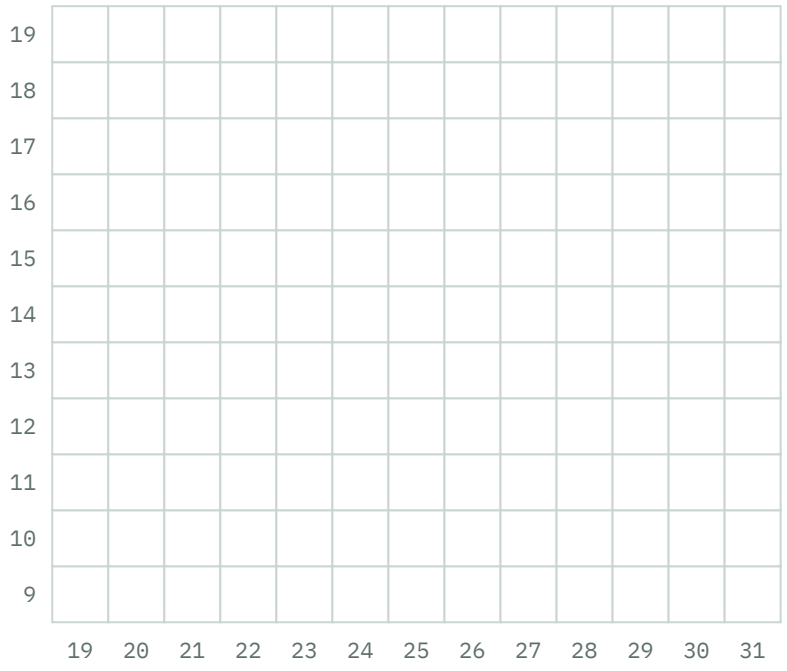
k	x	y	round(x)	round(y)
0				
1				
2				
3				
4				
5				
6				



Q9 BRESENHAM

Using Bresenham's algorithm, find the pixels from (20, 10) to (30, 18).

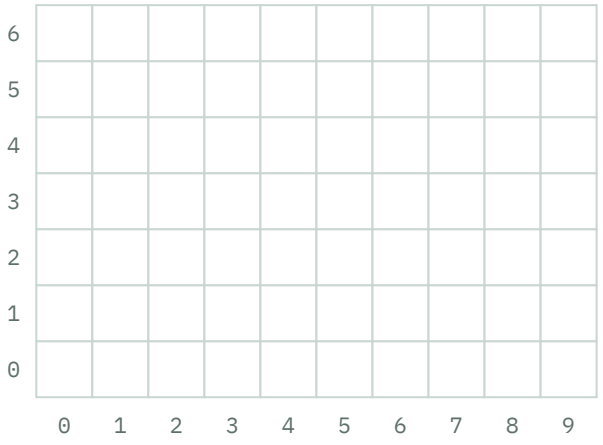
k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				



Q10 BRESENHAM

Using Bresenham's algorithm, find the pixels from (1, 1) to (8, 5).

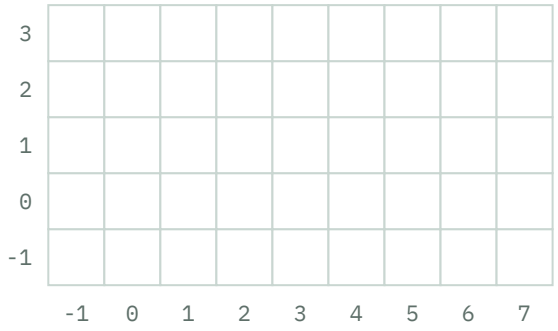
k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				



Q11 BRESENHAM

Using Bresenham's algorithm, find the pixels from (0, 0) to (6, 2).

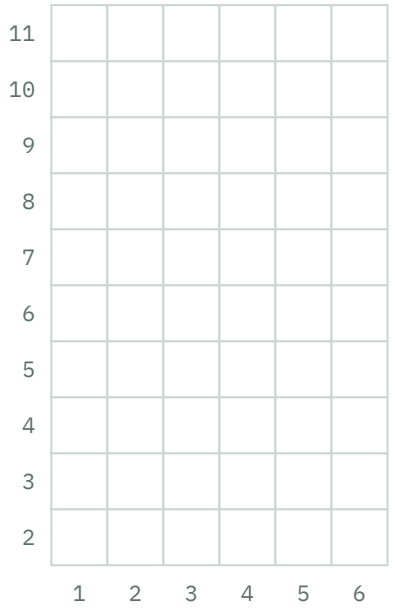
k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				



Q12 BRESENHAM · m > 1

Using Bresenham's algorithm, find the pixels from (2, 3) to (5, 10).

k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				



Q13 CIRCLE

Using Bresenham's circle algorithm, find the pixels of the first (top) octant of a circle centered at the origin with r = 8.

Q14 CIRCLE

Using Bresenham's circle algorithm, find the pixels of the first (top) octant for r = 10, then write the 8 symmetric points of (3, 10).